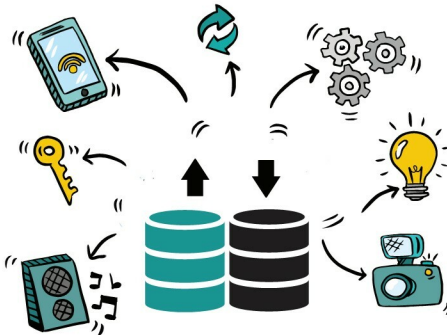


Open Source Databases that Work Best for IoT

The Internet of Things (IoT) generates vast amounts of data, including streaming data, time series data, RFID data, sensory data, etc. The efficient management of this data demands the use of a database. The very nature of IoT data requires a different type of database. Here are some databases that give very good results when used in conjunction with IoT.



The Internet of Things (IoT) can be regarded as a network in which various things are connected to each other through a common platform. Just visualise a scenario in which every device at home and the workplace is connected, and a world where the air-conditioning in a room automatically lowers its temperature when the outside temperature rises up, when the number of people in any public gathering is easily known, and when one's health parameters can be monitored on a daily basis. This is the possible impact of the Internet of Things.

The current state of the Internet of Things is very fragmented. There are different companies and organisations that are building their own platforms for either their customers or their individual needs. But a common platform on which all the devices, irrespective of their company, can be connected with each other via a user friendly interface, is still missing.

IoT devices are estimated to number in the trillions in the coming five years.

Is a database necessary for IoT?

The Internet of Things creates many tedious challenges,

especially in the field of database management systems, like integrating tons of voluminous data in real-time, processing events as they stream and dealing with the security of data. For instance, IoT based traffic sensors applied in smart cities would produce huge amounts of data on traffic in real-time.

Databases have a very important role to play in handling IoT data adequately. Therefore, along with a proper platform, the right database is equally important. As IoT operates across a diverse environment in the world, it becomes very challenging to choose an adequate database.

The factors that should be considered before choosing a database for IoT applications are:

- 1) Size, scale and indexing
- 2) Effectiveness while handling a huge amount of data
- 3) User-friendly schema
- 4) Portability
- 5) Query languages
- 6) Process modelling and transactions
- 7) Heterogeneity and integration
- 8) Time series aggregation
- 9) Archiving
- 10) Security and cost